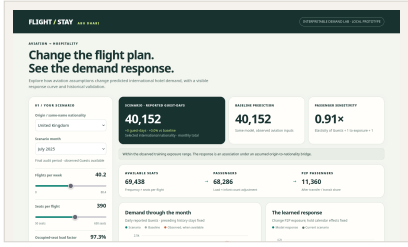
PROPOSAL / WORKING PROTOTYPE

Flight / Stay

A tool that links flights to hotel demand in Abu Dhabi.

EXECUTIVE SUMMARY

We built an offline simulator for the markets that already have flights. Change the flights, seats, load factor or connecting share. The simulator shows the change in hotel demand.
On held-back June–July 2025, adding flight data to a calendar-only model cut the monthly market error from **28.9%** to **23.1%**. We also measure which levers and markets move hotel demand most.



The prototype runs now. Slide 8 shows the full view.

Aaviskar Gautam, Srijan Sthapit (NYU Abu Dhabi) · Abdulrahman Alsharhan, Sarthak Malla (MBZUAI) · October 2026

Who we are and what we built

STUDENT TEAM

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CURRENT STATUS

- Our two notebooks run from start to end.
- We compared five readable models on a forward time split.
- The simulator is one HTML file. It runs with no server.
- The browser sliders and Python give the same numbers.

We want a first tool that a planner can question. We do not want a forecast that nobody can explain.

WHY THIS MATTERS

If flights change, what happens to hotel demand?

Airlines and hotels plan from different files. One seat can hold a transfer passenger, a returning resident or a hotel guest. So a direct seat-to-room number would be wrong.

QUESTIONS TO TEST

- What if the flights per week go up or down?
- What if an airline uses a larger aircraft?
- Which months and markets moved most in the past data?

The challenge asks for a tool that works, shows its steps and has a back-test. Source: ATP 2026 DCT Abu Dhabi challenge statement.

The two files count countries differently

WHAT THE FILES CONTAIN

- The flight file gives daily seats, passengers and connections.
- The hotel file gives daily *Guests* by nationality.
- The hotel file has 45 nationalities. 33 match a flight origin by name.
- The *Guests* numbers stop in July 2025.

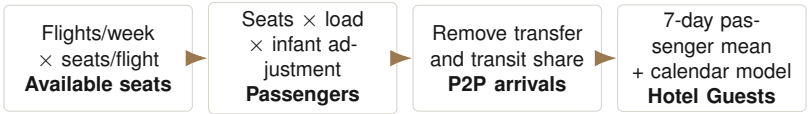
OUR STATED ASSUMPTIONS

- We link an origin to a nationality by name. This is an assumption, not a measurement.
- The other 12 nationalities get a calendar forecast. Their flight sliders stay off.
- We call the monthly sum of *Guests* reported guest-days. We cannot separate capture, length of stay and occupancy.

We drop the days with no hotel number. We start in 2023 because the 2022 flight rows are monthly, not daily.

How the simulator works

FIRST PLAIN ARITHMETIC, THEN A FITTED RESPONSE



INPUTS

Market, month, flights per week, seats per flight, load factor and transfer/transit share.

OUTPUTS

Predicted guest-days, the change from the real flights, the daily profile, the learned response and a warning for extreme inputs.

The fitted response holds the step from visitor to hotel guest. We cannot measure that step on its own.

We chose the simpler model on evidence

FIVE CANDIDATES

1. Calendar-only baseline
2. Proportional passenger chain
3. Monotonic regression **(selected)**
4. Monotonic GAM, 7-day exposure
5. Monotonic GAM, stay-kernel exposure

The two GAMs scored a little worse on the selection months. The extra curve gave no gain, so we left it out.

READABLE RESPONSE

$$\log(1 + \widehat{G}) = \alpha_m + \beta_m \log(1 + P_7) + \text{month} + \text{weekday} + \text{trend}$$
$$\beta_m \geq 0$$

Each matched market gets one coefficient. The coefficient cannot go below zero, so more passengers never lower demand. The seven-day mean lets an effect build up over days.

We chose the model on Jan–Apr 2025 only. If two models are within 1 point, we take the simpler one.

What the back-test shows

LOWER WMAPE IS BETTER

	Calendar	Flight model
Jan–Apr 2025 selection	31.9%	**25.0%**
Jun–Jul 2025 audit	35.0%	**27.7%**

WMAPE $= \sum |\text{prediction} - \text{observed}| / \sum \text{observed}$ over all nationality-days. We fit the models on Jan 2023–Dec 2024 after a 27-day warm-up. We keep May 2025 for the error bands.

OUR READING

7.2 points

Less error on the months we held back.

23.1% vs 28.9%

Error on monthly market totals: flight model vs calendar.

The daily total error is 6.5%, because one market's error can cancel another's. The test uses the flights that did operate. We changed no flights.

A working prototype, not just a diagram

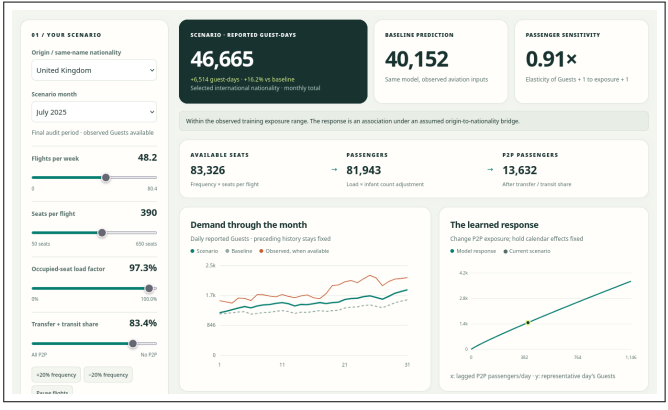
WHAT A PLANNER CAN DO NOW

1. Pick a market and a month.
2. Move a slider: flights/week, seats/flight, load factor or connecting share.
3. Compare with the real flights and read the passenger steps.
4. Watch for the warning past the training range.

EXAMPLE SHOWN

UK, July 2025, 20% more flights per week: **46,665** guest-days vs **40,152** with the real flights (**+16.2%**).

One HTML file, no network. The browser math matches Python.



The HTML prototype after the “+20% frequency” button.

What moves hotel demand, and what to do about it

SENSITIVITY: WHAT MOVES GUESTS MOST

- Flights per week, seats per flight and load factor act the same way: +10% in any one gives about $\beta_m \times 10\%$ more guests.
- Connecting share is the strongest lever per point. In UK/July 2025, 83% of arrivals connect onward. One point less means about 6% more local passengers.
- Markets differ more than 25 times: β is 1.9 for Qatar and Saudi Arabia, 0.9 for the UK and 0.07 for Lebanon.

WHAT PLANNERS CAN DO DIFFERENTLY

Focus route support on markets with high β and low connecting share, in their peak months. Test each case in the simulator before talks with airlines.

REAL-LIFE REQUIREMENTS

- Ask DCT what *Guests* counts, and for room counts to show occupancy.
- Link flight origin to nationality with approved aggregate data.
- Reload data often. Track error by market and season.
- Keep new routes off until evidence supports them.

The slider shows an association under the stated assumptions, not a proven causal effect of adding flights.

If shortlisted: test the hard part

DEMO AND TEST PLAN

1. Show three runs for one market and month: the baseline, 20% more flights, and a route cut.
2. Ask DCT to state the target definitions and the origin-to-nationality map.
3. Name the new test months and the real route changes first. Then check the error and the direction of the change.
4. Report the misses by market, by season and by how far the scenario goes past the data. Change the model only against months we have not used.

WHAT WE NEED TO IMPROVE

We need aggregate origin–nationality flows, a guest-night definition from DCT, event dates and room counts. With those we could test more of the seats-to-visitors-to-nights chain.

The prototype and the notebooks are ready for a live demo.

Sources: the ATP 2026 DCT Abu Dhabi challenge statement (challengeon.atrc.ae). The supplied DCT aviation and hotel files. Our domain analysis and model-comparison notebooks. We reproduced the results on 11 October 2026.